

Infrastructure planning

101-0509-10L (ECTS-6KP)

Appraising infrastructure modifications and adaptive plans

Project Sheet 2 (Weeks 3 and 4)

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Table of Contents

0	Purpose and scope of Phase 2.....	3
0.1	How to use the Exercise Sheets.....	4
1	Understanding the system model.....	5
1.1	What are we modelling? (tutorial/supporting task)	5
1.2	Input data and simulation setup (tutorial/supporting task)	6
1.3	Corridor and road network setup (tutorial/supporting task + mandatory task)	6
1.4	Baseline Passenger Demand & Desire Lines (tutorial/supporting task + mandatory task)	6
1.5	Baseline Modal Split (mandatory task + tutorial/supporting)	7
1.6	Dynamic (car) route assignment and congestion feedback (mandatory task + tutorial/supporting + advanced task)	7
1.7	Physical evaluation objectives (mandatory task).....	8
2	Model parameters	9
2.1	Core parameter overview (mandatory task)	9
2.2	Parameter playground: Year 1 vs. Year 40 (tutorial/supporting task).....	9
2.3	Sensitivity dashboard — parameters vs. objectives (tutorial/supporting task).....	9
3	Infrastructure stages	10
3.1	Representing physical infrastructure modifications (mandatory task)	10
3.2	Editing stage interventions (tutorial/supporting task + mandatory task).....	11
3.3	Comparing physical indicators across stages (mandatory task)	11
4	Deployment plans.....	12
4.1	Static plan definition (mandatory task)	12
4.2	Visualizing plans and their system pathways (mandatory task).....	13
4.3	Compare plan end states (tutorial/supporting task)	14
5	Summary: required outputs for Phase 02	14
6	Connection to SIA	15

0 Purpose and scope of Phase 2

During Phase 1, you defined the planning problem, relevant stakeholders, expected outcomes, and evaluation criteria for your case study. In Phase 2, you are now going to translate this conceptual understanding into a quantitative (transport) system model.

The objective is to understand:

- how the transport model represents the case study;
- how model parameters influence the calculated outcomes;
- how infrastructure modifications can be represented as planning stages;
- how these stages can be combined into different deployment plans/strategies.

The accompanying **02_system_modeling.ipynb** notebook guides you through these four components.

During this phase, you will be using a 4-step transport model. All model inputs are treated as **fixed assumptions, for now, and you will explore uncertainty-related and temporal aspects in future phases.**

During this phase, the transport model is primarily used as a set of **snapshots**. The main comparison concerns the state of the system in Year 1 and/or Year 40, under fixed assumptions and different infrastructure configurations.

The only temporal element introduced in this phase is the **deployment schedule of the infrastructure stages, in order to create so-called “plans”**.

0.1 How to use the Project Sheets

The present exercise sheet provides a short introduction to the course and specific information for the second phase of the semester (Weeks 3-4).

The previous (Weeks 1-2) and subsequent phases (Weeks 5–11) are supported by dedicated project sheets.

The project sheets combine conceptual tasks that help students develop the structure of their final report with practical analyses implemented in the accompanying **Jupyter Notebooks (.ipynb)** provided in the GitHub repository.

Throughout the exercise sheets, tasks are grouped into different categories to clarify what is expected from all groups and to provide opportunities for deeper exploration.

We distinguish between:

Supporting Tasks (optional)

Designed to help you understand the modelling framework, become familiar with the Jupyter notebooks, and develop an intuition for the rationale behind each analysis. These tasks do not directly produce outputs for the final report.

However, especially for students with limited experience in Python or quantitative modelling, we strongly recommend completing them, as they provide the foundation needed for the mandatory tasks.

Mandatory Task

Specifying core activities and deliverables that every group is expected to complete and include in the final report.

Advanced Task (optional)

Suggesting extensions for groups interested in exploring additional analyses, testing alternative assumptions or methods, or gaining a deeper understanding of the framework.

The exercises will **generally follow this structure**: first letting you explore and understand the modelling framework, then asking you to perform mandatory analyses, and finally giving you hints to further extend your approach. The exercises will not count as part of the grading. They provide an opportunity to get feedback on your work. You can submit your results (e.g., figures and tables) on Moodle and/or provide access to the lecturers and assistants to your Fork before the next Project sheet is released. You are expected to receive feedback within a week.

1 Understanding the system model

1.1 What are we modelling? (tutorial/supporting task)

Before modifying parameters, defining infrastructure stages and how these should be deployed over time, you must understand the system represented by the model, its spatial and temporal boundaries, and the assumptions connecting transport outputs to the Phase 1 evaluation criteria.

The supplied four-step transport model (FSM) represents peak-hour passenger trips (17:00–18:00) between Origin–Destination pairs within the Canton of Zurich. A conventional four-step model comprises:

1. Trip generation, which estimates the number of trips produced and attracted by each zone;
2. Trip distribution, which connects origins and destinations and balances the resulting OD matrix;
3. Mode choice, which distributes passenger demand across different modes (in our case: car, public transport with walking access, public transport with cycling access, cycling, and walking)
4. Traffic assignment, which loads road demand onto the network and estimates route flows, capacity utilisation, travel times, and congestion.

In this implementation, mode choice is probabilistic and is calculated with a Multinomial Logit model. Road assignment uses an iterative Method of Successive Averages (MSA) with capacity-dependent BPR travel times to approximate a static user equilibrium: travellers are, in a first iteration, assigned to the currently shortest routes, while congestion progressively changes route travel times until network conditions stabilise.

Congested road travel times are written back to the skim matrices and subsequently used to recalculate modal choice. Assignment and mode choice are repeated until both route flows and mode shares stabilise.

The principal inputs are the cantonal zone system, baseline passenger demand, modal travel time and distance skims, road network links and capacities, and calibrated mode choice parameters. The model also uses assumptions concerning fares, vehicle occupancy, e-bike use, emissions, perceived travel time, demand growth, and the spatial scope of interventions.

Infrastructure stages modify selected network and service attributes, before mode choice and impact indicators are calculated. The main quantitative outputs generally include passenger trips and modal shares, passenger-kilometres, vehicle-kilometres, travel times, road flows, volume-to-capacity ratios, congestion delay, and direct emissions.

Their interpretation depends on the selected spatial boundary, demand assumptions, parameter values, and aggregation method. In any case, some relevant effects remain outside the quantitative model (e.g. political feasibility or perceived landscape/public space quality), which require additional evidence through a qualitative assessment.

The things that you are asked to specify in further detail are stage-specific infrastructure effects, implementation plans/strategies that deploy different stages over time, and calculations of outputs such as modal shares, travel time, carbon emissions, passenger-kilometres, health effects, accident effects, and project costs.

The modelling pipeline used in this course does not reproduce every feature of the real transport system (although you can further expand the scope of the current modelling pipeline). However, it allows you to derive relevant information in order to take informed decisions about the infrastructure planning project you're confronted with and, as we will see in the next phases of the semester, test its performance under "deeply uncertain" future conditions.

You can find a more thorough explanation of how the model works on GitHub.

1.2 Input data and simulation setup (tutorial/supporting task)

Before running the model, you should verify the availability of the prepared FSM inputs (e.g. spatial zones and municipality attributes, baseline passenger and background demand, multimodal travel-time and distance skims, and more).

The preflight distinguishes between the inputs required for mode choice and those required for road-network analysis. No synthetic replacement should be interpreted as a valid baseline when required data are missing.

Tutorial/supporting task

Use the preflight cell and check whether all items are present and loaded correctly.

1.3 Corridor and road network setup (tutorial/supporting task + mandatory task)

The notebook shows, through a first network map, the complete cantonal road network. It represents transport supply rather than traffic conditions. You can inspect road class, free-flow speed, directional capacity, number of lanes, link length, and free-flow travel time. At this point, no traffic demand has been assigned and no congestion has been calculated.

However, for detailed assignment, the network is reduced to the case-study area, by using “gates” that allow for the simplification of the demand structure.

Mandatory Task

Select the relevant municipalities/zones for your case study and reduce the network. To do this, you can follow the methodology described in the notebook which allows for the creation of “gates”.

You can set a specific number of gates that should be used, and limit how close they can be to each other. By inspecting the total amount and distribution of candidate gates, you can adapt these values in order not to oversimplify the OD-matrix and risk losing relevant information. You can test multiple gate configurations and compare results by executing the cells that are already given to you in the notebook (after adapting them to your case study).

1.4 Baseline Passenger Demand & Desire Lines (tutorial/supporting task + mandatory task)

The cantonal Origin–Destination matrix represents passenger trips between the 1,223 FSM zones during the evening peak hour. Each cell identifies a directed movement from an origin zone to a destination zone.

Tutorial/supporting task

You can use the predefined cells in the notebook to visualise these patterns through maps with desire lines and/or through tabular representations of OD-matrices. In general, by leveraging both visualisation approaches, we advise you to:

- inspect municipality-to-municipality and zone-to-zone OD demand and desire lines – within your case-study area;
- distinguish **internal** → **internal**, **gate** → **internal**, **internal** → **gate**, and **gate** → **gate** (estimated pass-through) demand;

External origins and destinations are assigned to the nearest direction-compatible gate. Pass-through demand is assumption-dependent and should therefore be interpreted with particular care.

Mandatory Task

Validate the baseline passenger demand for your case-study area. Compare the main internal, inbound, outbound, and pass-through patterns with available independent evidence (e.g. observed transport data, official statistics, previous studies, CBA assumptions, or earlier course projects). Identify and document relevant discrepancies and their possible implications for subsequent analyses.

1.5 Baseline Modal Split (mandatory task + tutorial/supporting)

Before simulating infrastructure interventions, inspect the Stage-0 (baseline scenario, still with no interventions – which we will define in Part 3 of this exercise sheet/notebook) reference modal split. The model represents Car, Public Transport (access via cycling or walking), Cycling, and Walking.

Mandatory Task

Report both trip-based modal shares and Passenger-Kilometre (PKM) shares. In the Mehrspur example, PKM shares are filtered to trips longer than 5 km to provide a strategic indicator of longer-distance travel. Given the selected case study, consider if such approach is needed or not, and adjust the given value.

Calculate and visualise the baseline modal split using the predefined notebook cells:

- report total number of trips and trip-based shares for Car, Public Transport (decide yourself if aggregated or not), Cycling, and Walking;
- report the corresponding PKM shares;
- compare modal shares for the main OD relations relevant to your case study;
- interpret and explain the difference between trip-based and PKM-based modal shares
- perform the analysis again for a set of relevant OD-pairs.

Validate the resulting modal shares against available reference values. Where relevant, inspect differences between internal, inbound, outbound, and pass-through demand and document important discrepancies.

At this point, do not change the main parameter file. The purpose is to understand and verify the supplied reference model before adapting it in Part 2. If you change parameters, you should come back to this exercise and check whether substantial changes can be identified.

Tutorial/supporting task

Use the utility and parameter dashboard to explore how changes in baseline mode preferences affect your network and area. This dashboard is meant to help you achieve a good intuition of how the model works, without needing to inspect its details.

This is a controlled model-behaviour experiment, not an uncertainty analysis and not evidence that the tested value is realistic. We will perform more advanced sensitivity and uncertainty analyses later in the course.

Before continuing, we advise you to check (e.g. by adding a short check cell of your own) that modal probabilities sum to one for every Origin–Destination pair, aggregate modal shares sum to one, all probabilities lie between zero and one, and trips by mode sum to total demand.

1.6 Dynamic (car) route assignment and congestion feedback (mandatory task + tutorial/supporting + advanced task)

Car demand resulting from mode choice is assigned to the reduced road network together with fixed background traffic. The assignment uses All-or-Nothing shortest paths, the Method of Successive Averages (MSA), and BPR volume-delay functions to represent congestion.

The model first performs assignment with fixed car demand, allowing the route-assignment mechanism to be inspected independently. A second, coupled calculation feeds congested OD-specific car travel times back into mode choice, allowing travellers to change mode as congestion changes.

Tutorial/supporting task

Use the assignment dashboard to inspect:

- assigned flows and main traffic corridors;
- volume-to-capacity ratios;
- congested travel times and delay;
- link speed and capacity;
- the evolution of the MSA convergence diagnostic.

Use the convergence test to assess whether the selected iteration limit and road relative gap $\leq 2\%$ threshold provide a sufficiently stable result for the model. This threshold is a practical numerical diagnostic, not a formal proof of user equilibrium.

Mandatory Task

Run the coupled Stage-0 assignment and:

- identify the main congestion bottlenecks in your case-study area;
- compare the fixed-demand and congestion-feedback results;
- explain how congestion affects car travel times, modal choice, and network flows;
- document any important approximation or limitation affecting the interpretation of the assignment.

Before continuing, check that the assignment produces plausible flows, non-negative travel times and delays, and sufficiently stable results across the final MSA iterations.

Advanced Task (optional)

Use the assignment results together with the demand and modal-split analyses to assess whether Stage 0 credibly represents current conditions.

- compare modelled traffic volumes, congestion, speeds, travel times, congestion delay, and modal split with independent real-world evidence;
- ensure that model and reference data use compatible locations, directions, time periods, units, vehicle or passenger definitions, occupancy assumptions, and geographic boundaries;
- investigate discrepancies in demand, gate configuration, background traffic, occupancy, link attributes, mode-choice parameters, travel-time skims, and MSA convergence;
- where evidence justifies calibration, modify only clearly identified parameters and repeat the complete validation chain;
- document every adjustment, its evidence, and its effect without tuning the model to only one counter or indicator.

Keep calibration assumptions separate from infrastructure-stage modifications. Stage 0 should remain a transparent representation of current conditions.

1.7 Performance evaluation objectives (mandatory task)

Before comparing infrastructure stages, define the performance indicators used to evaluate the system. In the supplied notebook, the core indicators are:

- demand-weighted mode-choice journey time (min/trip; for PT including in-vehicle, access, egress and waiting time);
- road congestion delay (vehicle-hours per peak hour, from the car traffic assignment);
- car CO2 emissions (tonnes CO2 per peak hour; annual values also include rail electricity emissions);
- trips and modal shares in the corridor (passenger trips per peak hour; %).

Unlike the strategic modal-split indicator, these system-wide indicators are calculated over the full corridor. In particular, congestion and travel-time effects on shorter trips should not be excluded when they result from changes elsewhere in the network. Later notebook sections (2.3, 3.3 and 4.3) report the corresponding simulation-engine indicators: `avg_travel_time_min` (peak car/PT in-vehicle time plus car delay, min per motorised trip), `congestion_delay_hours` (passenger delay, person-hours), `co2_tonnes` (car and rail operating CO2) and `total_demand` (passenger trips).

Mandatory Task

Calculate the performance indicators for the Stage-0 Year-1 and Year-40 snapshots and:

- report each indicator with its unit;
- compare the two snapshots and explain the main differences;
- state whether a higher or lower value is preferable for each objective;
- relate the indicators to the evaluation criteria established in Phase 1.

You may add case-specific performance indicators where needed. Keep the objectives physical and transparent; monetary valuation and aggregation are addressed in later phases.

2 Model parameters

Model behaviour depends on assumptions concerning demand, travel behaviour, and transport-network performance. The notebook distinguishes in particular between:

- macro demand dynamics, such as cumulative demand growth by Year 40;
- behavioural sensitivities, such as mode affinities and utility parameters;
- network and service attributes, such as travel time, waiting time, transfer time, access/egress time, speed, and distance.

Additional parameters describe economic valuation, environmental impacts, and fixed engineering assumptions. Clearly distinguishing their roles is important when interpreting model results.

2.1 Core parameter overview (mandatory task)

The notebook provides an overview of the main parameters, their baseline values, units, spatial scope, location in the code, and direct role in the model.

Mandatory Task

Review the parameter overview and adapt the parameters needed to represent your case study. For every value that you modify or introduce, document:

- the parameter name, definition, value, and unit;
- its spatial scope and role in the model;
- the evidence source, calibration basis, or transparent reasoned assumption;

- the expected direction of its effect on relevant model outputs.

You are not expected to change every supplied parameter. Retain values that are appropriate for your case study and document important assumptions. Avoid introducing unexplained numerical constants directly in notebook cells.

2.2 Parameter playground: Year 1 vs. Year 40 (tutorial/supporting task)

Supporting Task (optional)

Use the parameter playground to explore how selected parameters affect the Stage-0 system in Year 1 and Year 40.

- vary cumulative demand growth and observe its effect on Year-40 demand and congestion;
- vary PT preference and transport-service parameters and inspect changes in modal split, CO2 emissions, travel time, and congestion;
- predict the expected direction of change before moving a slider;
- distinguish parameters that affect both snapshots from parameters that primarily affect the future snapshot.

Each selected setting is calculated with coupled MSA (mode choice and road assignment together) and compared with a nominal run; completed calculations are cached. Reset the parameters to their nominal values afterward. Slider ranges are diagnostic ranges and should not automatically be interpreted as realistic parameter values or uncertainty ranges.

2.3 Sensitivity dashboard — parameters vs. objectives (tutorial/supporting task)

Use the sensitivity dashboard to inspect how the physical objectives respond to changes in the main model parameters. The notebook performs a deterministic one-at-a-time sensitivity sweep: each parameter is varied while all others remain at their nominal values, with Stage 0 evaluated for Year 40. Each value is calculated with coupled mode choice and road assignment; the sweep is optional (RUN_PARAMETER_SWEEP = True) and can take substantially longer than a single simulation.

Supporting Task (optional)

Use the results to:

- identify parameters to which each objective is particularly sensitive;
- identify weak or negligible relationships;
- identify non-linear or unexpected responses;
- determine which assumptions require particularly careful evidence and specification.

This sensitivity sweep is intended as a first diagnostic step and does not constitute uncertainty analysis. Because each parameter is varied on its own, it does not capture interactions between parameters.

Where conclusions depend on combinations of parameters, test a small number of combined settings in the parameter playground.

The results should help you specify the model parameters and, in later phases, identify which uncertainties require particular attention when structuring the uncertainty analysis.

3 Infrastructure stages

3.1 Representing infrastructure modifications (mandatory task)

A stage is a distinct physical and operational state of the transport system. It is not a year or a deployment plan. Each stage bundles a coordinated set of infrastructure interventions and their corresponding model effects.

Mandatory Task

Define two independently deployable infrastructure packages in `code/stages.py` (`PACKAGES['stations']` and `PACKAGES['tunnel']` in the MehrSpur example). The model assembles the following configurations from them:

- Stage 0: reference/no-intervention system (assembled automatically);
- Stage 1: first package only (e.g. the station package);
- Stage 2: second, independently deployable package only (e.g. the tunnel package); the combined configuration with both packages (state 3), including any `COMBINED_EFFECTS`, is assembled automatically.

When defining each stage, specify:

- physical scope: what is actually constructed or modified;
- spatial scope: the OD relations, zones, stations, or road links affected;
- magnitude: the numerical effect of each intervention and its evidence base;
- costs: relevant capital and operating costs;
- dependencies and timing: physical requirements and plausible deliverability;
- effects outside the model: relevant impacts that cannot be represented quantitatively.

The supplied model represents interventions differently by mode. Public-transport and bicycle interventions modify existing OD skim relations; mobility-hub interventions modify PT access, egress, waiting, and transfer components; road-capacity interventions act on individual road-network links and propagate through the assignment model. New PT or bicycle connections that do not exist in the baseline skims cannot be created simply by reducing existing travel times.

Be explicit about the spatial selector used for each intervention. An intervention does not automatically apply corridor-wide.

Each configuration is evaluated as a complete alternative relative to the baseline skims, not as a sequential update of another stage. The combined configuration contains the effects of both packages once, plus any `COMBINED_EFFECTS`: minute-based railway savings (section time, headway) and capacity increases add up, while OD-based percentage interventions of both packages are applied one after another.

For every numerical modification, use a defensible source or calculation rather than an arbitrary percentage. After defining the stages, verify that each stage contains the intended interventions and that the resulting transport effects move in a plausible direction.

3.2 Editing stage interventions (tutorial/supporting task + mandatory task)

You can use the notebook editor to inspect and test provisional PT, bicycle, mobility-hub, and directed road-link interventions. The editor changes in-memory working copies only; definitive and reproducible interventions must be transferred to `code/stages.py`.

Otherwise, you can directly adjust these values in the source files.

Mandatory Task

For every edited intervention:

- identify and justify each affected OD pair, hub area, or directed road link;
- verify its spatial selector and modified attribute on the editor map;
- for road interventions, verify the before/after values in the road-link audit table;
- distinguish the represented model effect from relevant impacts outside the model boundary;
- transfer accepted definitions to code/stages.py. Save the script, restart the kernel and Run All again to reload updates.

The notebook variables EDITED_STAGE_SPECS, ROAD_STAGE_CORRIDORS, and ROAD_STAGE_LINK_AUDITS are provisional and are discarded when the editor is reset or the kernel is restarted.

3.3 Comparing performance indicators across stages (mandatory task)

The stages must be compared under identical non-stage assumptions. Year 1 compares complete stage packages at reference demand; Year 40 compares the same packages under the deterministic demand-growth assumption.

The Year-40 comparison is conditional: it shows what the model calculates if the specified fixed assumptions hold and the selected infrastructure stage is active. It should not be described as a probability, uncertainty range, or robust forecast.

Mandatory Task

Compare all defined infrastructure stages using the same Year-1 and Year-40 snapshot assumptions.

- Report car, public-transport, bicycle, and walking trip and PKM shares for every stage.
- Compare Year-1 and Year-40 peak-hour trips by mode and the additional performance indicators.
- Use the stage-summary tables and the accompanying bar-chart panels to compare the same values visually.
- For Stage 0 → Stage 1, Stage 0 → Stage 2 and Stage 0 → Both Stages combined, explain the full causal chain from physical modification to parameter changes, utilities or performance, modal split, and impacts.
- Identify trade-offs rather than assuming that the largest stage improves every objective.

Supporting Task (optional)

Interpret the effect of each transition between stages.

- List the infrastructure modifications introduced by the transition.
- List the exact parameters changed by those modifications.
- Quantify the change in modal split and the selected objectives.
- Explain why the model produces the observed result.
- Identify at least one important effect missing from the quantitative table.
- Discuss possible trade-offs, such as road-space reallocation improving cycling conditions while reducing car capacity or speed.

Investigate counter-intuitive results. Do not simply report them without checking units, stage definitions, and the causal mechanism.

Before continuing, verify that Stage 0 reproduces the intended reference system, every stage changes only the intended parameters, unchanged parameters remain identical, dependencies are respected, modal shares remain complete and non-negative, and repeated runs of the same stage under identical assumptions produce identical results.

Before moving to Phase 3, save the final package definitions in `code/stages.py` (Notebook 02, Section 3.4). Temporary changes made in the interactive editor are not used by Notebook 03, which trains the transport surrogate from the saved definitions.

Optionally, Notebook 02, Section 3.5 counts the passengers crossing a selected counting section and calibrates a supplementary external flow (`EXTERNAL_FLOW`) against an observed count (tutorial/supporting task). If a counting section and external flow is not being used, these should be disabled.

4 Deployment plans

4.1 Static plan definition (mandatory task)

A plan describes how infrastructure stages are deployed. It connects physical states to decisions.

The same stages can be combined into different plans. For example, one policy may retain Stage 0 throughout the planning horizon, another may construct the most extensive stage immediately, and another may construct Stage 1 first and Stage 2 at a predetermined later year. An adaptive plan may state a measurable condition under which the stages would be deployed.

Notebook 02 defines and compares the deterministic baseline and static plans. Staged and adaptive plans, including considerations regarding trigger timing and robustness, are examined in later phases.

Mandatory Task

Define the deterministic deployment plans in `code/adaptive_planning.py` by editing the `PLAN_DEFINITIONS` dictionary near the top of the file. In Notebook 02, adapt the supplied baseline and static plans to your case study.

For each plan, specify:

- a clear name and description;
- the packages open from Year 1 (`initial_packages`);
- stations and tunnel set to None for these static plans (no later openings);
- the relevant package costs (set in each package's appraisal block in `code/stages.py`);
- stage dependencies and important irreversibilities;
- a concise description of the plan logic;
- the stage active in Year 40;
- the plan key included in `PLAN_SUBSET`.

Include a no-intervention baseline and an upfront static plan. Staged and adaptive plans are examined in Notebook 04. After saving `adaptive_planning.py`, rerun the setup and plan summary cells.

4.2 Visualizing plans and their system pathways (tutorial/supporting task)

A metro style pathway map represents the active state over the planning horizon. The horizontal axis shows time, the vertical axis shows the infrastructure stage, horizontal segments show the stage that is active, and vertical movements mark deployment decisions. Fixed decisions and conditional rules should use different markers.

Supporting Task (optional)

Generate a metro-style pathway map from the selected plan definitions.

- Show the planning year on the horizontal axis and the infrastructure stage on the vertical axis.
- Use one clearly labelled line for each plan.
- Use horizontal segments to show the active stage; later stage transitions may be added in subsequent phases.
- If additional plans are shown, distinguish fixed decisions from conditional rules using the supplied markers.
- Use descriptive stage labels, a legend, and a title that can be understood without reading the code.
- Ensure that the map makes the initial stage, final stage, and any fixed timing or conditional decisions clear.

Explain the map briefly below the figure and distinguish committed from conditional decisions. An illustrative year attached to a conditional rule is not a forecast.

4.3 Compare plan end states (Mandatory Task)

At this phase, policy comparisons are limited to the physical stage reached in the fixed Year-40 snapshot. Policies that end in the same stage should produce the same Year-40 transport result when all other assumptions are identical, even if their earlier deployment paths differ. Their accumulated benefits, lifecycle costs, and discounted values may differ because timing matters, but that appraisal is addressed in later notebooks.

Mandatory Task

Use the policy end-state table as a consistency check.

- Compare the Year-40 stage, modal split, travel time, congestion delay, CO₂, total demand, 40-year cost and other indicators of each plan.
- State which differences can only be evaluated when the full deployment pathway, costs, and accumulated effects are appraised in later phases.

5 Summary: required outputs for Phase 02

Mandatory Task

By the end of Phase 2, every group should have completed the mandatory Notebook 02 activities and retained the following material:

- a concise explanation of the system boundary, four-step structure, calculation chain, and limitations;
- a completed FSM input preflight and data-readiness check;
- a validated baseline OD matrix and desire-line analysis covering internal, inbound, outbound, and pass-through demand;
- a validated Stage-0 modal split reported by trips and PKM, including relevant OD relations;
- a coupled Stage-0 assignment comparison documenting congestion feedback and convergence;
- the four performance indicators for Stage 0 in Year 1 and Year 40, with units and preferred directions;
- a documented parameter register with values, units, spatial scope, model role, evidence, and expected effects;
- two reproducible package definitions in code/stages.py (four configurations) plus validated intervention selectors and road-link audits where applicable;

- a Year-1 and Year-40 comparison of all stages using modal split, trips, performance indicators, and the notebook's bar-chart panels;
- baseline and static plan definitions in code/adaptive_planning.py;
- a Year-40 plan using the notebook's table and bar-chart panels.

The playground, sensitivity sweep, calibration extension, and other explicitly labelled supporting or advanced activities remain optional. Staged and adaptive plans are introduced in Notebook 04.

6 Connection to SIA

The activities carried out during Phase 2 support the transition from the initial problem definition to the development and comparison of possible infrastructure solutions within the strategic planning process.

In particular, the work relates to the following planning activities:

- translating functional, economic, environmental, and societal objectives into measurable system characteristics;
- defining the system boundary, calculation methods, indicators, and assumptions used to evaluate alternatives;
- developing and structuring possible infrastructure modifications and variants;
- identifying the expected effects and interactions of the proposed modifications;
- documenting investment, operating, and performance implications;
- developing staged implementation concepts and alternative deployment strategies;
- preparing a transparent quantitative basis for the later uncertainty analysis and appraisal.

The deterministic model developed in this phase forms the quantitative foundation for the uncertainty modelling, adaptive-planning analysis, and final appraisal conducted in the subsequent phases of the semester.